

ABDOULAYE ALHASSANE DIALLO

New York, NY | diallo.abdoulaye@columbia.edu | (347) 992-5926

abdoulayediallo.com | github.com/Kienda | linkedin.com/in/abdoulaye-alhassane-diallo

EDUCATION

Columbia University — B.A. in Computer Science, New York, NY

Expected May 2028

Jack Kent Cooke Scholar (national merit scholarship, <1% selection rate) | Dean's List

Coursework: Data Structures & Algorithms, CS Theory, Advanced Programming (C/C++), Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, React Native, Next.js, Node.js, Express.js, FastAPI, Qt, Tailwind CSS

Data & AI: PostgreSQL, SQLite, REST APIs, RAG pipelines, ChromaDB, sentence-transformers, LLM APIs (Groq, IBM Watson)

Infrastructure: AWS, Nginx, Cloudflare (DNS/SSL), Vercel, GitHub Actions (CI/CD), Linux (Debian/Ubuntu), Git

Spoken Languages: English, French, Arabic, Fula

EXPERIENCE

Software Engineer & Tech Lead — Heuvia (heuvia.com), Remote

2025 – Present

- Lead a 5-person team (coordinating 3 developers) that built a multi-tenant Learning Management System for colleges in Guinea, supporting student, professor, and administrator roles
- Designed the system architecture: multi-tenant subdomain routing (Nginx reverse proxy on a Debian VPS + Cloudflare DNS), a 4-role permission system, and curriculum-based enrollment on a React / React Native / Node.js / PostgreSQL stack
- Configured server infrastructure and transactional SMTP email delivery; own production deployment, code review, and task coordination across web and mobile (development complete, preparing first institutional rollout)

IRT Support Specialist — Research Foundation of CUNY, New York, NY

Jun – Aug 2025

- Resolved 200+ weekly Microsoft 365 / Active Directory requests (provisioning, MFA, access control) with a 99% resolution rate; deployed and configured 100+ Windows machines across campus systems

IT Manager — Guinea High Tech, Conakry, Guinea

2015 – 2022

- Scaled technical operations from 1 to 10+ team members, delivering end-to-end IT solutions to 250+ businesses; automated system deployment and standardized configurations across client environments

Web Development Intern — Identix Guinée, Conakry, Guinea

Jul – Nov 2018

- Developed web application features and contributed to relational database schema design using Laravel (PHP) and Bootstrap

PROJECTS

Tontin — Digital Tontine / Group Savings Platform | *FastAPI, Python, PostgreSQL, AWS (Ubuntu)*

tontin.net

- Building a full-stack fintech app for trust-based rotating savings circles: wallets, group contribution pots, goal-based savings vaults, and a member trust-scoring system (MVP live in demo)
- Built a FastAPI REST backend with full CRUD over a relational database and a web frontend; self-deployed on an AWS EC2 Ubuntu instance managed from the terminal

The Unofficial Guide — RAG Question-Answering System | *Python, ChromaDB, sentence-transformers, Groq*

github.com/Kienda

- Built an end-to-end retrieval-augmented generation system over Columbia CS professor reviews (376 reviews, 394 chunks, all-MiniLM-L6-v2 in ChromaDB); engineered grounded generation with source citations, out-of-scope refusal, and retrieval evaluation

Alkule — Full-Stack Language Learning Platform | *Next.js, TypeScript, Node.js, PostgreSQL, Firebase*

alkule.com

- Designed, built, and deployed a live platform for learning the ADLaM (Fulani) script with typing practice, courses, and a media library; integrated Firebase auth/storage and PostgreSQL, deployed via Railway with Cloudflare DNS/SSL and a custom domain

ColumbiaCrisis — Student Housing Review Platform | *React, Express.js, PostgreSQL, REST APIs*

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- Built a full-stack housing-review platform for Columbia students with secure email verification, review management, and a relational schema behind a responsive React interface

StudentSuccess — Academic Guidance Platform | *Node.js, Express.js, PostgreSQL, JWT*

github.com/Kienda

- Built a full-stack platform with role-based authentication (JWT, bcrypt), a relational PostgreSQL schema, and server-rendered personalized student dashboards

College Navigator — Desktop Academic Tracker (Honors Project) | *C++, Qt, SQLite*

- Developed a cross-platform C++/Qt desktop app tracking academic milestones with SQLite persistence and an object-oriented architecture separating UI, logic, and data layers